25/01/2023
Evening

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2023 (Online) Phase-1

(Physics, Chemistry and Mathematics)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) The Test Booklet consists of 90 questions. The maximum marks are 300.

(3) There are **three** parts in the question paper consisting of **Physics, Chemistry** and **Mathematics** having 30 questions in each part of equal weightage. Each part (subject) has two sections.

(i) **Section-A:** This section contains 20 multiple choice questions which have only one correct answer. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer.

(ii) **Section-B:** This section contains 10 questions. In Section-B, attempt any **five questions out of 10.** The answer to each of the questions is a numerical value. Each question carries **4 marks** for correct answer and **–1 mark** for wrong answer. For Section-B, the answer should be rounded off to the nearest integer.

PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Every planet revolves around the sun in an elliptical orbit:

   A. The force acting on a planet is inversely proportional to square of distance from sun.

   B. Force acting on planet is inversely proportional to product of the masses of the planet and the sun.

   C. The Centripetal force acting on the planet is directed away from the sun.

   D. The square of time period of revolution of planet around sun is directly proportional to cube of semi-major axis of elliptical orbit.

   Choose the correct answer from the options given below:

   (1) C and D only (2) B and C only

   (3) A and C only (4) A and D only

Answer (4)

Sol. A is correct because $F = \frac{Gm_1m_2}{r^2}$

D is correct because $T^2 \propto r^3$

$\therefore$ A and D are correct

2. Match List I with List II

	List I		List II
A.	Isothermal Process	I.	Work done by the gas decreases internal energy
B.	Adiabatic Process	II.	No change in internal energy
C.	Isochoric Process	III.	The heat absorbed goes partly to increase internal energy and partly to do work
D.	Isobaric Process	IV.	No work is done on or by the gas

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV (2) A-I, B-II, C-IV, D-III

(3) A-II, B-I, C-IV, D-III (4) A-II, B-I, C-III, D-IV

Answer (3)

Sol. A. Isothermal process $\rightarrow$ II $(\Delta U = 0)$

B. Adiabatic process $\rightarrow$ I $(\Delta Q = 0)$

C. Isochoric process $\rightarrow$ IV $(w = 0)$

D. Isobaric process $\rightarrow$ III $(\Delta Q = \Delta U + w)$

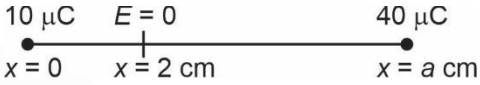
3. A point charge of 10 μC is placed at the origin. At what location on the X-axis should a point charge of 40 μC be placed so that the net electric field is zero at x = 2 cm on the X-axis?

   (1) x = 8 cm (2) x = 6 cm

   (3) x = – 4 cm (4) x = 4 cm

Answer (2)

Sol.



$\therefore \quad E_{x=2\text{ cm}} = 0$

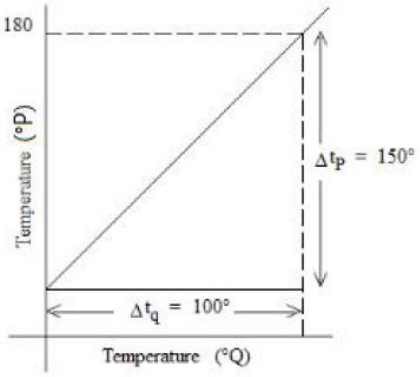
$$\frac{1}{4\pi\varepsilon_0}\frac{(10\ \mu C)}{(2\ cm)^2} = \frac{1}{4\pi\varepsilon_0}\frac{(40\ \mu C)}{[(a-2)\ cm]^2}$$

$$\Rightarrow \left(\frac{a-2}{2}\right)^2 = 4$$

$$\Rightarrow \frac{a-2}{2} = 2$$

$$\boxed{a = 6\ cm}$$

4. The graph between two temperature scales P and Q is shown in the figure. Between upper fixed point and lower fixed point there are 150 equal divisions of scale P and 100 divisions on scale Q. The relationship for conversion between the two scales is given by:



(1) $\frac{t_P}{180} = \frac{t_Q - 40}{100}$ (2) $\frac{t_Q}{100} = \frac{t_P - 30}{150}$

(3) $\frac{t_P}{100} = \frac{t_Q - 180}{150}$ (4) $\frac{t_P}{150} = \frac{t_P - 180}{150}$

Answer (2)

Sol. From graph, we can say that.

$$\frac{t_P - 30}{150} = \frac{t_Q - 0}{100}$$

5. Given below are two statements:

Statement I: Stopping potential in photoelectric effect does not depend on the power of the light source.

Statement II: For a given metal, the maximum kinetic energy of the photoelectron depends on the wavelength of the incident light.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Both Statement I and Statement II are incorrect

(4) Statement I is incorrect but statement II is correct

Answer (1)

Sol. **Statement I** is correct as stopping potential is independent of power of light used.

Statement II is correct as maximum kinetic energy of photoelectron depends on wavelength of light.

6. The light rays from an object have been reflected towards an observer from a standard flat mirror, the image observed by the observer are: -

A. Real

B. Erect

C. Smaller in size than object

D. Laterally inverted

Choose the *most appropriate* answer from the options given below:

(1) B and D Only (2) A and D Only

(3) A, C and D Only (4) B and C Only

Answer (1)

Sol. The image will be **erect** and **laterally inverted**.

7. Two objects are projected with same velocity '*u*' however at different angles α and β with the horizontal. If $\alpha + \beta = 90°$, the ratio of horizontal range of the first object to the 2nd object will be :

(1) 1 : 2 (2) 4 : 1

(3) 2 : 1 (4) 1 : 1

Answer (4)

Sol. Both the objects will have the same range because, $\alpha + \beta = 90°$. *i.e.*, α, β are complementary angles.

8. Match List I with List II

List I		List II	
A.	Troposphere	I.	Approximate 65-75 km over Earth's surface
B.	E-Part of Stratosphere	II.	Approximate 300 km over Earth's surface
C.	F_2-Part of Thermosphere	III.	Approximate 10 km over Earth's surface
D.	D-Part of Stratosphere	IV.	Approximate 100 km over Earth's surface

Choose the correct answer from the options given below:

(1) A-III, B-II, C-I, D-IV (2) A-I, B-II, C-IV, D-III

(3) A-I, B-IV, C-III, D-II (4) A-III, B-IV, C-II, D-I

Answer (4)

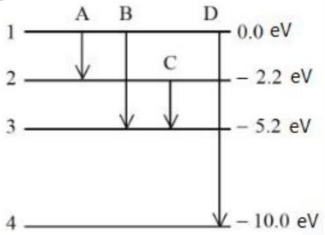
Sol. $\rightarrow$ 10 km over Earth's surface – Troposphere

$\rightarrow$ 100 km over Earth's surface – E-part of stratosphere

$\rightarrow$ 300 km over Earth's surface – F_2-part of thermosphere

$\rightarrow$ 65-75 km over Earth's surface – D-part of stratosphere

9. The energy levels of an atom is shown in figure.



Which one of these transitions will result in the emission of a photon of wavelength 124.1 nm?

Given ($h = 6.62 \times 10^{-34}$ Js)

(1) D

(2) B

(3) C

(4) A

Answer (1)

Sol. $\because \lambda = \frac{hc}{\Delta E}$

$$\lambda = \frac{1240}{10}\text{nm}$$

≈ 124 nm.

$\therefore$ D is the transition required.

10. For a moving coil galvanometer, the deflection in the coil is 0.05 rad when a current of 10 mA is passed through it. If the torsional constant of suspension wire is 4.0×10^{-5} N m rad^{-1}, the magnetic field is 0.01 T and the number of turns in the coil is 200, the area of each turn (in cm^2) is :

(1) 1.0 (2) 0.5

(3) 2.0 (4) 1.5

Answer (1)

Sol. $\because \theta = \left(\frac{NBA}{K}\right)I$

$$A = \frac{\theta K}{NBI}$$

$$= \frac{0.05 \times 4 \times 10^{-5}}{(200) \times (0.01) \times \left(10 \times 10^{-3}\right)}$$

$= 1$ cm^2

11. A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If time taken by particle to go from $x = 0$ to $\frac{A}{2}$ is 2 s; then time taken by particle in going from $x = \frac{A}{2}$ to A is

(1) 3 s (2) 2 s

(3) 1.5 s (4) 4 s

Answer (4)

Sol. $x = A\sin(\omega t)$

$$x = \frac{A}{2} = A\sin(\omega t)$$

$$\frac{1}{2} = \sin(\omega t)$$

$$t = \left(\frac{\pi}{6\omega}\right) = 2$$

$$\boxed{\frac{\pi}{\omega} = 12 \text{ sec}}$$

$x = A = A\sin(\omega t)$

$$\omega t = \left(\frac{\pi}{2}\right)$$

$$t = \left(\frac{\pi}{2\omega}\right) = 6 \text{ second}$$

time = 6 – 2 = 4 seconds

12. A wire of length 1 m moving with velocity 8 m/s at right angles to a magnetic field of 2 T. The magnitude of induced emf, between the ends of wire will be_____

(1) 20 V (2) 16 V

(3) 8 V (4) 12 V

Answer (2)

Sol. $\varepsilon = BVl$

$= 2 \times 8 \times 1$

$= 16$ V

13. The distance travelled by a particle is related to time t as $x = 4t^2$. The velocity of the particle at $t = 5$ s is:

(1) 8 ms^{-1} (2) 20 ms^{-1}

(3) 40 ms^{-1} (4) 25 ms^{-1}

Answer (3)

Sol. $x = 4t^2$

$$\frac{dx}{dt} = 8t$$

v at $t = 5$ s = 40 m/s

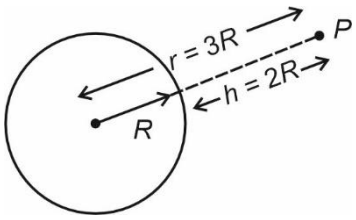
14. A body of mass is taken from earth surface to the height h equal to twice the radius of earth (R_e), the increase in potential energy will be:
(g = acceleration due to gravity on the surface of Earth)

(1) $\frac{1}{3}mgR_e$ (2) $3mgR_e$

(3) $\frac{2}{3}mgR_e$ (4) $\frac{1}{2}mgR_e$

Answer (3)

Sol.



$$V_{surface} = -\left(\frac{GMm}{R_e}\right)$$

$$V_p = -\frac{GMm}{3R_e}$$

$$\Delta V = \frac{GMm}{R_e}\left(1 - \frac{1}{3}\right)$$

$$= \frac{2}{3}\frac{GMm}{\left(R_e^2\right)} \times R_e$$

$$\Delta V = \frac{2}{3}mgR_e$$

15. Match List I with List II

	List I		List II
A.	Young's Modulus (Y)	I.	$[ML^{-1}T^{-1}]$
B.	Co-efficient of Viscosity (η)	II.	$[ML^2T^{-1}]$
C.	Planck's Constant (h)	III.	$[ML^{-1}T^{-2}]$
D.	Work Function (φ)	IV.	$[ML^2T^{-2}]$

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV (2) A-III, B-I, C-II, D-IV

(3) A-II, B-III, C-IV, D-I (4) A-I, B-III, C-IV, D-II

Answer (2)

Sol. $(A)[Y] = \left[\frac{MLT^{-2}}{L^2}\right] = \left[ML^{-1}T^{-2}\right]$ ...(III)

$$\frac{F}{A} = \eta\left(\frac{dV}{dY}\right)$$

$$(B)[\eta] = \left[\frac{MLT^{-2}\times L}{L^2\times LT^{-1}}\right] = \left[ML^{-1}T^{-1}\right] \quad ...(I)$$

$h\nu = E = [ML^2T^{-2}]$

(C) $[h] = [ML^2T^{-1}]$ ...II

(D) $\phi = [ML^2T^{-2}]$ ...IV

16. Match List I with List II

	List I		List II
A.	Gauss's Law in Electrostatics	I.	$\oint \vec{E}\cdot\vec{dl} = -\frac{d\phi_B}{dt}$
B.	Faraday's Law	II.	$\oint \vec{B}\cdot\vec{dA} = 0$
C.	Gauss's Law in Magnetism	III.	$\oint \vec{B}\cdot\vec{dl} = \mu_0 i_c + \mu_0 \in_0 \frac{d\phi_E}{dt}$
D.	Ampere-Maxwell Law	IV.	$\oint \vec{E}\cdot\vec{ds} = \frac{q}{\in_0}$

Choose the correct answer from the options given below.

(1) A-II, B-III, C-IV, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-I, C-II, D-III

(4) A-I, B-II, C-III, D-IV

Answer (3)

Sol. Gauss's law $\oint \vec{E}\cdot\vec{ds} = \frac{q}{\in_0}$ (A → IV)

Faraday's law $\oint \vec{E}\cdot\vec{dl} = -\frac{d\phi_B}{dt}$ (B → I)

Gauss's law in magnetism $\oint \vec{B}\cdot\vec{dA} = 0$ (C → II)

Ampere's-Maxwell law $\oint \vec{B}\cdot\vec{dl} = \mu_0 i_c + \mu_0 \in_0 \frac{d\phi_E}{dt}$

(D → III)

17. According to law of equipartition of energy the molar specific heat of a diatomic gas at constant volume where the molecule has one additional vibrational mode is

(1) $\frac{3}{2}R$ (2) $\frac{7}{2}R$

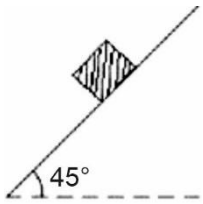
(3) $\frac{9}{2}R$ (4) $\frac{5}{2}R$

Answer (2)

Sol. According to the equipartition of energy degree of freedom of diatomic gas is $f = 7$, (2 degree of freedom is added for every vibrational mode) So,

$$C_v = \frac{f}{2}R = \frac{7R}{2}$$

18. Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to



(1) 0.60 (2) 0.25

(3) 0.50 (4) 0.33

Answer (4)

Sol. Force required to push

$$F_1 = mg\sin\theta + \mu mg\cos\theta = \frac{mg}{\sqrt{2}}(1+\mu)$$

Force required to prevent from sliding

$$F_2 = (mg\sin\theta - \mu mg\cos\theta) = \frac{mg}{\sqrt{2}}(1-\mu)$$

Given $F_1 = 2F_2$

$$1 + \mu = 2(1 - \mu)$$

$$\mu = \frac{1}{3} = 0.33$$

19. **Statement I:** When a Si sample is doped with Boron, it becomes *P* type and when doped by Arsenic it becomes N-type semiconductor such that P-type has excess holes and N-type has excess electrons.

Statement II: When such P-type and N-type semi-conductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In the light of above statements, choose the ***most appropriate*** answer from the options given below.

(1) Both statement I and statement II are correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

Answer (2)

Sol. Statement I is correct but in statement II we cannot detect the current through ammeter thus the statement II is incorrect.

20. The resistance of a wire is 5 Ω. It's new resistance in ohm if stretched to 5 times of its's original length will be

(1) 25 (2) 5

(3) 125 (4) 625

Answer (3)

Sol. $R' = n^2R$

$= 5^2 \times 5\ \Omega$

$= 125\ \Omega$

SECTION - B

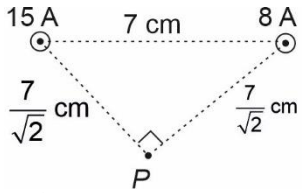
Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

21. Two long parallel wires carrying currents 8 A and 15 A in opposite directions are placed at a distance of 7 cm from each other. A point *P* is at equidistant from both the wires such that the lines joining the point *P* to the wires are perpendicular to each other. The magnitude of magnetic field at *P* is _____ × 10^{-6} T.

(Given : $\sqrt{2} = 1.4$)

Answer (68)

Sol. From question



$$\vec{B}_{15A} = \frac{\mu_0 \times 15}{2\pi\left(\frac{7}{\sqrt{2}}\text{ cm}\right)}, \qquad \vec{B}_{8A} = \frac{\mu_0 \times 8}{2\pi\left(\frac{7}{\sqrt{2}}\right)\text{cm}}$$

$\vec{B}_{15A}$ & $\vec{B}_{8A}$ are perpendicular to each other.

Hence $\vec{B}_P = \vec{B}_{15A} + \vec{B}_{8A}$

$$= \sqrt{(B_{15A})^2 + (B_{8A})^2}$$

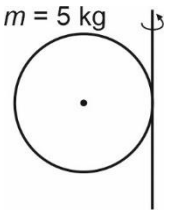
$$= \frac{\mu_0}{2\pi\left(\frac{7}{\sqrt{2}}\right)\text{cm}}\sqrt{18^2 + 8^2} = \frac{\mu_0 17}{2\pi \times \left(\frac{7}{\sqrt{2}}\right) \times 10^{-2}}$$

$$= \frac{4\pi \times 17 \times 10^{-7}}{2\pi \times \frac{7}{\sqrt{2}} \times 10^{-2}} = 68 \times 10^{-6}$$

22. If a solid sphere of mass 5 kg and a disc of mass 4 kg have the same radius. Then the ratio of moment of inertia of the disc about a tangent in its plane to the moment of inertia of the sphere about its tangent will be $\frac{x}{7}$. Then the value of *x* is _____

Answer (5)

Sol. Solid sphere



$I_{tangent} = I_{cm} + mR^2$

$$= \frac{2}{5}mR^2 + mR^2 = \frac{7}{5}mR^2$$

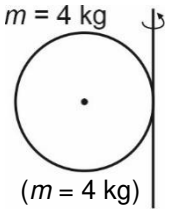
$= 7R^2$ $(m = 5\text{ kg})$

$I_{disc} = I_{cm} + mR^2$

$= \frac{mR^2}{4} + mR^2$

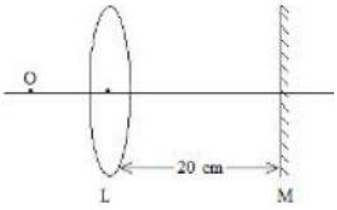
$= \frac{5}{4}mR^2$

$= 5\ R^2$



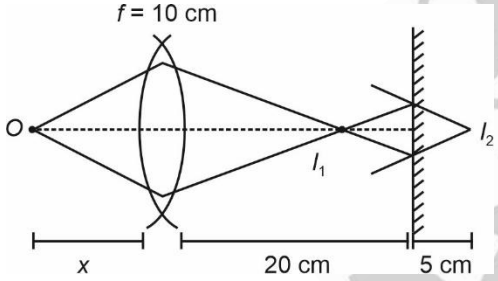
$\frac{I_{disc}}{I_{tangent}} = \frac{5}{7}$

23. An object is placed on the principal axis of convex lens of focal length 10 cm as shown. A plane mirror is placed on the other side of lens at a distance of 20 cm. The image produced by the plane mirror is 5 cm inside the mirror. The distance of the object from the lens is _____ cm



Answer (30)

Sol.



From diagram

I_1 is image formed by lens and I_2 is image formed by mirror.

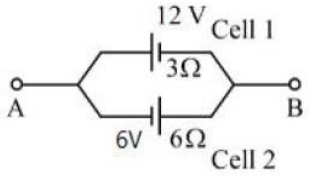
Location of I_1 and I_2 from mirror will be equal = 5 cm

Hence I_1 = 15 cm from lens

From $\frac{1}{v} - \frac{1}{u} = \frac{1}{10}$; $u = -x,\ v = 15$

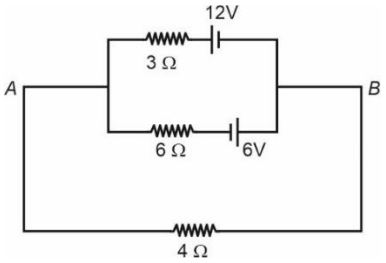
$\frac{1}{x} = \frac{1}{10} - \frac{1}{15} \Rightarrow x = 30$ cm

24. Two cells are connected between points A and B as shown. Cell 1 has emf of 12 V and internal resistance of 3 Ω. Cell 2 has emf of 6 V and internal resistance of 6 Ω. An external resistor R of 4 Ω is connected across A and B. The current flowing through R will be _____A.



Answer (1)

Sol.



KCL at A gives

$\frac{6 - V_A}{4} + \frac{0 - V_A}{6} + \frac{18 - V_A}{3} = 0$

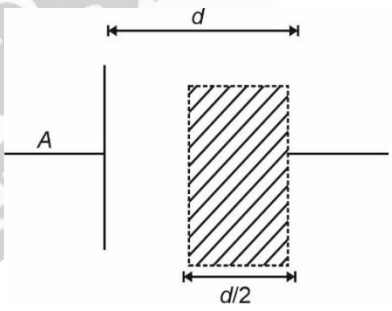
$V_A = 10$

So current through 4 Ω = $\frac{10 - 6}{4} = 1$ A

25. A capacitor has capacitance 5 μF when it's parallel plates are separated by air medium of thickness d. A slab of material of dielectric constant 1.5 having area equal to that of plates but thickness $\frac{d}{2}$ is inserted between the plates. Capacitance of the capacitor in the presence of slab will be _____μF

Answer (6)

Sol.



$C = \frac{\varepsilon_0 A}{d} K$

When completely air filled

$C = 5\mu F = \frac{\varepsilon_0 A}{d}$ ...(1)

When half filled with $K = 1.5$

$\frac{1}{C_{eq}} = \frac{\frac{d}{2}}{\varepsilon_0 A} + \frac{\frac{d}{2}}{\varepsilon_0 AK}$

$C_{eq} = \left(\frac{2K}{K+1}\right)\frac{\varepsilon_0 A}{d}$ ...(2)

From (1) & (2)

$C_{eq} = \left(\frac{2 \times 1.5}{1.5 + 1}\right) 5\ \mu F = 6\ \mu F$

26. A train blowing a whistle of frequency 320 Hz approaches an observer standing on the platform at a speed of 66 m/s. The frequency observed by the observer will be (given speed of sound = 330 ms^{-1}) Hz.

Answer (400)

Sol. $f = f_0\left(\frac{v}{v - v_s}\right)$

$f = 320\left(\frac{330}{330 - 66}\right)$

$= 320 \times \frac{330}{264}$

= 400 Hz.

27. A nucleus disintegrates into two smaller parts, which have their velocities in the ratio 3:2. The ratio of their nuclear sizes will be $\left(\frac{x}{3}\right)^{\frac{1}{3}}$. The value of 'x' is

Answer (02)

Sol. $m_1 v_1 = m_2 v_2$

$\Rightarrow \left(\frac{m_1}{m_2}\right) = \frac{v_2}{v_1} = \left(\frac{2}{3}\right)$

$m \propto A$

$\frac{A_1}{A_2} = \left(\frac{2}{3}\right)$

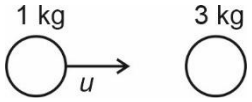
$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{1/3} = \left(\frac{2}{3}\right)^{1/3}$

$x = 2$

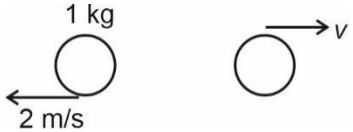
28. A body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg. After collision, the smaller body reverses its direction of motion and moves with a speed of 2 m/s. The initial speed of the smaller body before collision is _____ ms^{-1}.

Answer (04)

Sol. Before collision



After collision



Momentum conservation

$u + 0 = 3v - 2$

$\boxed{3v - u = 2}$ …(1)

also,

$\frac{v + 2}{u} = 1 \Rightarrow v + 2 = u$

$\boxed{u - v = 2}$ …(2)

Adding (1) and (2)

$2v = 4$

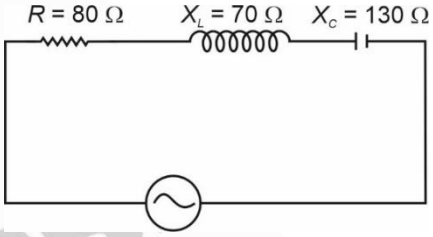
$v = 2$ m/s

$\boxed{u = 4 m/s}$

29. A series LCR circuit is connected to an AC source of 220 V, 50 Hz. The circuit contains a resistance R = 80 Ω, an inductor of inductive reactance X_L = 70 Ω, and a capacitor of capacitive reactance X_C = 130 Ω. The power factor of circuit is $\frac{x}{10}$. The value of x is:

Answer (08)

Sol.



Power factor = $\cos\phi = \frac{R}{Z} = \frac{80}{\sqrt{80^2 + 60^2}}$

$\frac{8}{10} = \frac{x}{10} \Rightarrow \boxed{x = 8}$

30. A spherical drop of liquid splits into 1000 identical spherical drops. If u_i is the surface energy of the original drop and u_f is the total surface energy of the resulting drops, the (ignoring evaporation), $\frac{u_f}{u_i} = \left(\frac{10}{x}\right)$. Then value of *x* is _____ :

Answer (01)

Sol. $\frac{u_f}{u_i} = \frac{\text{Area of final drop}}{\text{Area of Initial drop}}$

$\frac{u_f}{u_i} = \frac{1000 \times 4\pi r_f^2}{4\pi r_i^2} = \frac{1000(r_f^2)}{(r_i^2)}$

$1000 \times \frac{4}{3}\pi r_f^3 = \frac{4}{3}\pi r_i^3$

$r_i = 10 r_f$

$\frac{u_f}{u_i} = \frac{1000 r_f^2}{100 r_f^2} = 10$

$\frac{10}{x} = 10 \Rightarrow x = 1$

CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

31. Which of the following represents the correct order of metallic character of the given elements?

(1) Be < Si < K < Mg (2) K < Mg < Be < Si

(3) Be < Si < Mg < K (4) Si < Be < Mg < K

Answer (4)

Sol. Metallic character of an element is directly proportional to its electropositivity. Of the given elements silicon is least electro positive and potassium is most electropositive whereas beryllium and magnesium have intermediate values in the increasing order. Therefore, correct order of metallic character is Si < Be < Mg < K.

32. Match List-I with List-II.

	LIST-I (Amines)		**LIST-II (pK_b)**
A.	Aniline	I.	3.25
B.	Ethanamine	II.	3.00
C.	N-Ethylethanamine	III.	9.38
D.	N, N-Diethylethanamine	IV.	3.29

Choose the correct answer from the options given below.

(1) A-III, B-II, C-I, D-IV (2) A-III, B-II, C-IV, D-I

(3) A-I, B-IV, C-II, D-III (4) A-III, B-IV, C-II, D-I

Answer (4)

Sol. Aromatic amines are less basic than aliphatic amines. Among given aliphatic amines, 2° amine is most basic, followed by 3° amine and 1° amine. Therefore the correct basic strength (K_b) order of the given amines is

$$\underset{(C)}{CH_3CH_2NHCH_2CH_3} > \underset{(D)}{(CH_3CH_2)_3N} > \underset{(B)}{CH_3CH_2NH_2} > \underset{(A)}{C_6H_5NH_2}$$

The pK_b order of the given amines will be just the opposite of their basic strength order. The correct matching is

A – III, B – IV, C – II, D – I

33. Given below are two statements, one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : Carbon forms two important oxides – CO and CO_2. CO is neutral whereas CO_2 is acidic in nature.

Reason R : CO_2 can combine with water in a limited way to form carbonic acid, while CO is sparingly soluble in water.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both A and R are correct and R is the correct explanation of A

(2) A is not correct but R is correct

(3) A is correct but R is not correct

(4) Both A and R are correct but R is NOT the correct explanation of A

Answer (1)

Sol. Carbon monoxide is neutral and CO_2 is acidic in nature because with the increase in oxidation state of carbon acidic strength increases. So, Assertion is correct.

CO_2 combines with water to form carbonic acid while CO is sparingly soluble in water. So, Reason is also correct and is the correct explanation of Assertion.

34. **Statement I :** Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.

Statement II : The crossed arrow of the dipole moment symbolized the direction of the shift of charges in the molecules.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is incorrect but Statement II is correct

(2) Both Statement I and Statement II are incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is correct but Statement II is incorrect